

WASP-104b

R-2485

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_180326 · created 2026-08-24T00:46:59+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458203.6473 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.044
Transit depth (Rp/Rs)^2	2.14 +/- 1.29 [%]
Orbital Inclination (inc)	87.59 +/- 2.88
Airmass coefficient 1 (a1)	1.59 +/- 0.96
Airmass coefficient 2 (a2)	-0.54 +/- 1.01
Scatter in the residuals of the lightcurve fit is	52.24 %
Best Comparison Star	#1 - [139, 137]
Optimal Aperture	1.7
Optimal Annulus	4.0
Transit Duration (day)	0.061 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.146 ± 0.044 vs 0.1214 (deviation 0.56σ)
Duration fit vs published	0.061 d vs 0.0742 d (deviation 1.65σ)
Mid-time O–C	-0.1 min
Depth SNR	3.3
Residual scatter	52.24 %

Light curve

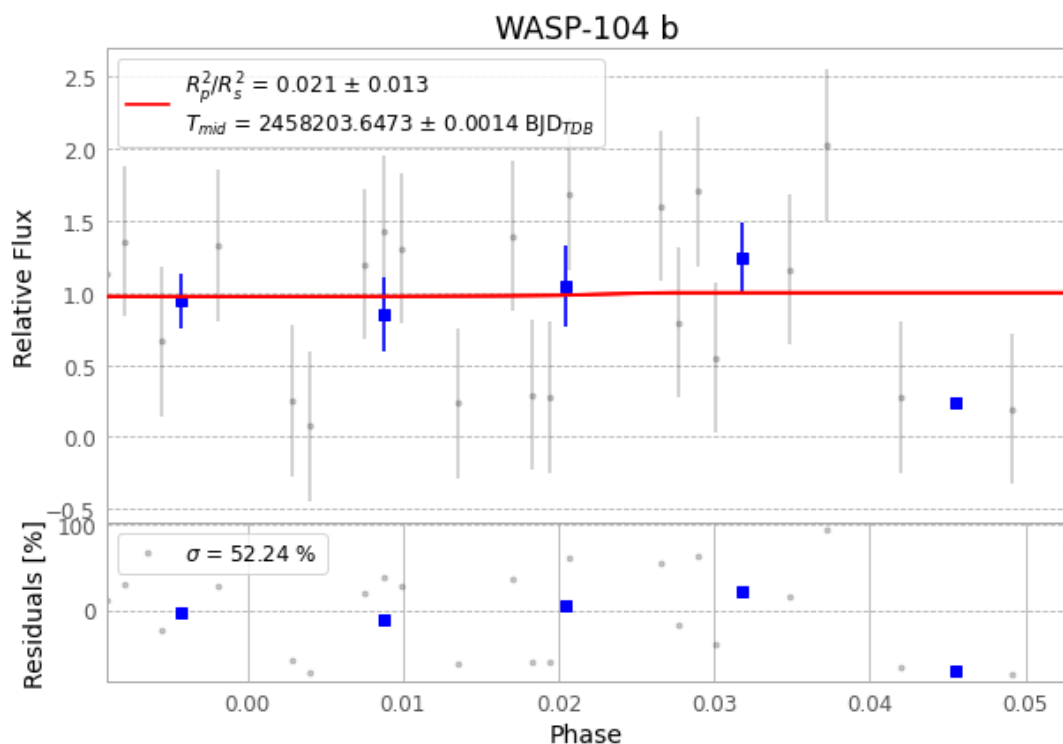


Figure 1 — FinalLightCurve_WASP-104 b_2018-03-25.png (SHA-256 6022c2c664ad7e27...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 183], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 61699a2227a896a1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), WASP-104180326024812.FITS → WASP-104180326053612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-180326.log (6a1b9e8dbc7c...), FinalLightCurve_WASP-104 b_2018-03-25.png (6022c2c664ad...), FinalLightCurve_WASP-104 b_2018-03-25.csv (6656af618745...)

Compute resources

Wall time 102.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 00:46Z.